PERFORMANCE TESTING

LegalEase AI

Date	29/09/2026
Team ID	SWTID - 2026 - 4944
Project Name	LegalEase AI
Maximum Marks	3 Marks

Performance Testing Document

PERFORMANCE TESTING

Project Name: LegalEase – AI-Powered Legal Document Generator

Testing Date: 29 September 2026

Maximum Marks: 5 Marks

1. Testing Overview

Field	Details
Testing Tool Used	Postman (API performance / response-time testing approach)
Type of Testing	Load Testing and Response-Time Testing
Target Module / API	FastAPI POST /generate endpoint
Test Environment	Local development environment – FastAPI backend with Gemini AI integration
Test Date	29 September 2026

2. Test Scenarios

The scenarios below are derived from LegalEase's documented workflow: submitting document type, parties, terms and effective date to the /generate endpoint and receiving AI-generated legal content.

S.No	Test Scenario / Description	Virtual Users	Duration (sec)	Expected Outcome
1	Single-user Employment Contract generation through /generate	1	60	Successful response with generated legal content
2	Concurrent NDA generation requests through /generate	5	60	Stable API responses with no request errors
3	Concurrent Lease Agreement generation requests	10	120	Service remains responsive under expected load
4	Peak-load mixed document generation (Contract/NDA/Lease)	20	120	Acceptable response time and error rate

3. Performance Test Results

Note: The values in this table are documented test-result targets/sample observations for the project report. The supplied LegalEase documentation does not contain measured JMeter/Postman/Locust result logs or screenshots, so these values should not be presented as independently verified benchmark measurements.

S.No	Metric	Target Value	Actual / Recorded Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.8 seconds	Pass	Within target for the documented test scenario
2	Response Time (Max)	< 5 seconds	4.2 seconds	Pass	Peak response remains below target
3	Throughput (Req/sec)	Stable under test load	Approx. 2.4 req/sec	Pass	Adequate for the sample load scenario
4	Error Rate	< 1%	0.5%	Pass	Below specified threshold
5	CPU Utilization	< 80%	68%	Pass	Within target
6	Memory Utilization	< 80%	71%	Pass	Within target

4. Observations & Analysis

• The primary performance target is the FastAPI POST /generate endpoint, which accepts document_type, parties, terms and dates and routes the request to GeminiDocumentGenerator.

• LegalEase uses Streamlit as the frontend, FastAPI as the backend, Gemini as the AI generation core, and formatting modules for TXT, DOCX and PDF output.

• Under the documented sample load, the recorded values remain below the template's response-time, error-rate, CPU and memory thresholds.

• AI generation time can vary because the /generate workflow depends on the external Gemini API. Therefore, repeated testing should be performed with the same environment, input size and model configuration when comparing results.

• The project documentation specifically requires end-to-end testing with different input combinations and verification of document generation, preview, editing and downloads.

• For a formal performance report, actual tool-generated evidence such as Postman/JMeter/Locust graphs, timestamps, request counts and server-resource measurements should be attached.

5. Screenshots / Evidence

Evidence to attach for final submission:

1	Postman/JMeter/Locust test execution screenshot for POST /generate
2	Response-time and request-count report
3	Error-rate result
4	CPU and memory utilization screenshot
5	Successful generated document response / API response screenshot

6. Conclusion

The LegalEase performance-testing plan focuses on the /generate API and the end-to-end AI document-generation workflow. The project architecture is designed for real-time responsiveness and local deployment, while the supplied testing template defines response-time, throughput, error-rate, CPU and memory metrics. Actual benchmark evidence should be captured from the selected performance-testing tool before treating the results as verified measurements.